

4. Experiments

Here we evaluate our approach across a broad range of vision and vision-language tasks. We provide extensive ablations on 29 tasks over the design choices in Section 4.2, and compare with state-of-the-art approaches on popular zero-shot benchmarks in Section 4.3. Finally, we present an alternate approach to do few-shot classification with prompt-based initialization in Section 4.4.

4.1. Experimental setup

Training datasets. We use a 2.1B English caption subset of the LAION-5B dataset [66]. Prior to training, we filter out sample pairs with NSFW images, toxic words in the text, or images with a watermark probability larger than 0.5, following [69]. This leaves us with 1.98B images, which we refer to throughout the paper as the LAION-2B dataset. Additionally, we explore training our models on a collection of Public Multimodal Datasets (PMD) from [70]. PMD contains training splits of various public datasets. After downloading⁵ the data we are left with 63M (vs. 70M reported in [70]) image-text pairs due to missing samples and SBU Captions [56] (originally in PMD) going offline.

Training details. For our model architecture, we closely follow CLIP by Radford *et al.* [62]. We utilize Vision Transformers (ViT) [17] for images and Text Transformers [75] for captions. We experiment with 3 different architectures, denoted as B/32, B/16, and L/14, where 32, 16, and 14 denote the input image patch size. See the supplementary for architecture details. For distillation and fine-tuning experiments, we utilize the public SWAG-ViT models [71], pre-trained with weak supervision from hashtags.

We use the Adam [33] optimizer with a decoupled weight decay [48] and a cosine learning rate schedule [47]. The input image size is 224×224 pixels. To accelerate training and save memory, we use mixed-precision training [51]. All hyperparameters are presented in the supplementary. They are selected by training B/32 on a small scale setup, and reused for all architectures. For objects and attributes classifiers, we found that scaling the learning rate by 10.0 and weight decay by 0.01 gave better results. We train our models on 4B, 8B, 16B, and 32B total samples. For ViT-L/14, we further train the model at a higher 336px resolution for 400M samples, denoting this model as L/14@336. We trained L/14 for 6 days on 512 A100 GPUs with 16B processed samples for a total of 7.4×10^4 GPU hours.

Evaluation benchmarks. We evaluate our models on a zero-shot benchmark of 29 tasks: (i) 17 image classification, (ii) 10 cross-modal retrieval, (iii) 2 visual question answering. Dataset details are presented in the supplement.

⁵Downloaded following huggingface.co/datasets/facebook/pmd.

Table 1. Evaluating effect of using LAION-2B subset filtered on complexity (C), actions (A), and text-spotting (T). CLIP denotes filtering pairs with CLIP score below 0.35. Evaluation performed on ViT-B/32 model architecture trained for 4B processed samples.

#	Filter				Size	IN	COCO		Flickr	
	CLIP	C	A	T			T2I	I2T	T2I	I2T
1					1.98B	60.8	33.7	52.1	59.3	77.7
2	✓				440M	52.5	29.8	46.1	54.8	72.0
3		✓			1.71B	60.8	33.9	52.5	60.8	77.8
4		✓	✓		642M	58.7	35.9	53.8	64.3	82.0
5		✓	✓	✓	438M	61.5	37.6	55.9	66.5	83.2

4.2. Ablations on zero-shot benchmarks

In this section, we ablate our three pretraining contributions: dataset filtering, distillation from objects and attributes predictions, and, hard negative contrastive objective. Ablations are performed over zero-shot Accuracy@1 on the ImageNet1K [65] (IN) validation set, text-to-image (T2I) and image-to-text (I2T) zero-shot Recall@1 on the COCO [60] and Flickr [59] test sets. We also report the change in accuracy (%) over 29 zero-shot tasks between our model and baselines. For a fair comparison, we train all approaches presented in this section (including baselines).

Effect of dataset filtering. We apply our filters, as well as filtering based on CLIP [62] alignment score (<0.35), and ablate the baseline performance, without distillation or hard negative contrastive training, in Table 1 for ViT-B/32 model architecture. All models see 4B total samples during training, while the number of unique samples drops after each filtering step. Complexity filter (C) in row (3) reduces the dataset size by around 270M, while slightly increasing image-text alignment as observed on T2I task. Next, action filter (A) in row (4) reduces the size by more than 1B, while it has a large benefit in aligning complex text. However, as expected, it hurts performance on object-centric ImageNet. Finally, text-spotting (T) filter in row (5) boosts alignment across the board, due to the fact that it removes the need to learn a bimodal visual representation of the text. We also compare with filtering based on CLIP score in row (2), which was selected such that the dataset size is comparable to ours, and show that it is too strict and removes plenty of useful training pairs, thus hurting the performance. Finally, LAION-CAT, with only **22%** of the original dataset size, significantly boosts image-text zero-shot performance. We also observed that gains hold as we train for longer training schedules. See the supplementary for details.

Effect of distillation approach. To understand the effect of direct distillation from a pre-trained SWAG-ViT visual encoder [71], we investigate two baseline approaches: (1) *Embedding distillation (ED)* borrows from SimSiam [9] and uses an auxiliary negative cosine similarity loss between the image representation from the student visual encoder and the pre-trained SWAG model.

(2) *Distribution distillation (DD)* borrows ideas from momentum distillation in ALBEF [41] and computes the cross-modal similarities between the SWAG image representation and the student text representation and uses them as soft-labels for student image representation and text alignment. The soft-labels are linearly combined with the hard 0 – 1 labels before applying the InfoNCE [55] loss.

A comparison of our distillation from predicted concepts (CD) with the aforementioned distillation approaches is presented in Table 2 (upper section). Note that for a fair comparison, we do not use our hard-negative contrastive loss for these experiments. Our distillation approach performs the best, even though it has virtually no training overhead as the predicted concepts are pre-computed, while, *e.g.*, ED is 60% slower with an 8% increase in GPU memory due to the need of running an additional copy of the vision tower. One could pre-compute embeddings for ED and DD as well, but that increases dataset size by 1.2TB and creates a data loading bottleneck, while our pre-computed predictions take only 32.6GB additional storage space when saving the top-10 predictions (see supplementary). We additionally show that our approach is robust to the number of top- k predictions used, details in the supplementary.

One could also use an external unimodal image model and fine-tune it on the image-text alignment task instead of using distillation. We follow [89] and explore three fine-tuning options as baselines: (i) locked-image tuning (LiT) where the image encoder is locked, and only the text encoder is trained, (ii) fine-tuning (FT) where the image encoder is trained with a learning rate scaled by 0.01 compared to the text encoder, (iii) fine-tuning with delay (FT-delay) where the image encoder is locked for half of the pre-training epochs following (i), and then fine-tuned for the rest following (ii). Results of these setups are ablated in Table 2 (lower section). LiT *vs.* FT is a trade-off between strong performance on image recognition tasks (as measured with ImageNet1K) and better image-text alignment (as measured by COCO and Flickr). Locking the image encoder makes the alignment very hard to achieve, but fine-tuning it hurts its original image recognition power. On the other hand, we show that our concept distillation is the best of both worlds, it surpasses LiT or FT in 4 out of 5 metrics. Another drawback of FT is that it requires the same architecture in the final setup, while CD can be effortlessly combined with any architecture or training setup, by using stored predictions as metadata. To conclude, unlike related approaches, our proposed distillation: (i) has almost no cost at training, (ii) is architecture agnostic, (iii) improves both image recognition and complex image-text alignment.

Effect of hard negative contrastive training. We present the ablation when using hard negative contrastive objective (HN-NCE) in Table 3. Performance suggests that using the newly proposed loss is beneficial compared to the vanilla

Table 2. Evaluating effect of using different initialization or distillation approaches. Evaluation performed on ViT-B/16 model architecture trained for 16B processed samples on LAION-CAT. Init: Initialization with random or SWAG-B/16 weights. ED: Embedding distillation. DD: Distribution distillation. LiT: Locked image tuning. FT: Fine-tuning. FT-delay: Locked image tuning for 50% followed by fine-tuning for the rest. CD: Our concept distillation using teacher-predicted objects and attributes.

Init	Method	SWAG (teacher)	IN	COCO		Flickr	
				T2I	I2T	T2I	I2T
Random	Baseline	—	68.7	42.8	60.5	72.8	89.7
	ED	B/16	69.2	42.6	59.4	72.8	86.8
	DD	B/16	68.6	41.8	57.4	71.7	87.0
	CD (ours)	B/16	71.0	42.8	59.5	72.3	86.5
	CD (ours)	H/14	72.3	43.4	60.4	73.8	87.6
SWAG	LiT	—	73.0	32.5	50.6	60.8	79.6
	FT	—	71.2	43.1	60.3	73.1	87.7
	FT-delay	—	72.0	42.7	60.7	72.5	86.2

Table 3. Evaluating effect of using hard negative contrastive loss. Evaluation performed on ViT-B/16 model architecture trained for 16B processed samples on LAION-CAT. CD: Our concept distillation using SWAG-H/14 predicted objects and attributes. HN: Our proposed hard negative contrastive loss.

#	Method		IN	COCO		Flickr	
	CD	HN		T2I	I2T	T2I	I2T
1			68.7	42.8	60.5	72.8	89.7
2	✓		72.3	43.4	60.4	73.8	87.6
3	✓	✓	72.0	43.7	62.0	73.2	89.5

InfoNCE, and that its positive effects are complementary to the gains from the proposed distillation from objects and attributes predictions. Please see the supplementary for ablations on the effect of the hyperparameters α and β .

Effect when pre-training on PMD. Finally, we analyze our proposed recipes when training visual-language models on a much smaller dataset, *i.e.* PMD with 63M training samples. Results are shown in Table 4. All contributions improve the performance over baseline significantly, hence we conclude that using the proposed pipeline is very beneficial in low-resource training regimes⁶. Note that, the PMD dataset contains COCO and Flickr training samples, hence, it is not strictly zero-shot evaluation. For that reason, we do not compare our models trained on PMD dataset with state-of-the-art models in the following section. However, we believe these strong findings will motivate usage of our approach on smaller and cleaner datasets, as well.

Zero-shot benchmarks. We denote model trained with our proposed concept distillation and hard-negative loss as DiHT. To showcase our model’s performance in more detail, we report our DiHT-B/16 trained on LAION-CAT with 438M samples *vs.* CLIP-B/16 baseline trained by us on LAION-2B with 2B samples in Figure 4. Additionally, we report DiHT-B/16 *vs.* CLIP-B/16 baseline, where both

⁶PMD is smaller and relatively much cleaner dataset compared to LAION. Hence, we observed that our filtering step is not needed for it.

Table 4. Evaluating effect when pre-training on PMD using our approaches. Evaluation performed on ViT-B/32 and ViT-B/16 models trained for 4B processed samples. CD: Our concept distillation using SWAG-H/14 predicted objects (-O) and attributes (-A). HN: Our proposed hard negative contrastive loss.

Arch.	#	Method			IN	COCO		Flickr	
		CD-O	CD-A	HN		T2I	I2T	T2I	I2T
B/32	1				49.0	28.9	50.2	62.0	80.3
	2	✓			57.8	32.2	54.0	65.6	85.7
	3	✓	✓		59.7	34.4	55.7	68.3	87.8
	4	✓	✓	✓	62.4	37.3	60.4	71.8	89.9
B/16	5				54.6	33.1	55.7	67.4	85.5
	6	✓			65.5	37.4	59.9	72.4	88.7
	7	✓	✓	✓	67.8	42.7	65.5	77.6	92.5

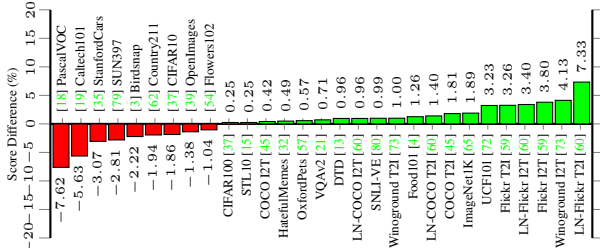


Figure 4. DiHT-B/16 trained on LAION-CAT with 438M samples vs. CLIP-B/16 trained on LAION-2B with 2B samples. Both models trained by us with 32B total processed samples.

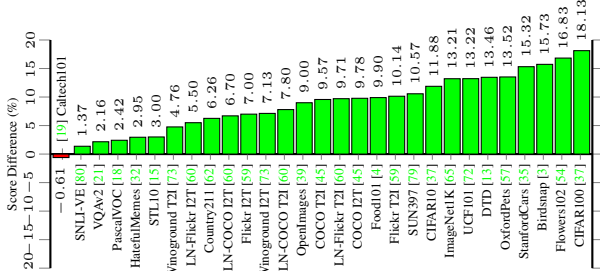


Figure 5. DiHT-B/16 vs. CLIP-B/16. Both models trained by us on PMD with 63M images and 4B total processed samples.

models are trained on PMD dataset with 63M samples in Figure 5. When trained on LAION-CAT or LAION-2B, respectively, DiHT wins on 20 out of 29 benchmark tasks. Impressively, when trained on PMD, DiHT wins on 28 out of 29 benchmark tasks, usually with a very large margin.

Robustness to distribution shift. We evaluate the ImageNet-related robustness on different datasets in Table 5, for DiHT-B/16 vs. CLIP-B/16 baseline, trained by us on PMD and LAION datasets. Our proposed approach improves on robustness over vanilla CLIP, in some cases by a significant margin, e.g. ImageNet-A and ObjectNet.

4.3. Comparison with zero-shot state of the art

We compare our DiHT models against state-of-the-art dual-encoder models in Table 6. Given that all models use different architectures, input image resolutions, training databases, and number of processed samples at training, we outline those details in the table, for easier comparison.

Table 5. ImageNet robustness performance on ViT-B/16 models.

Method	Train Data	#D	IN	IN-V2	IN-A	IN-R	IN-Sketch	ObjectNet
CLIP	PMD	63M	54.6	47.9	35.5	56.3	30.8	38.5
DiHT	PMD	63M	67.8	61.5	54.3	74.4	44.7	52.8
CLIP	LAION-2B	1.98B	70.3	62.7	39.3	81.0	57.1	56.2
DiHT	LAION-CAT	438M	72.2	64.3	49.2	85.1	58.3	62.3

Our approach is most similar to CLIP [62] and OpenCLIP [27], and has same training complexity and inference complexity. We outperform models with same architecture by substantial margins, even when our training dataset is much smaller. Our best models DiHT-L/14 and DiHT-L/14@336 trained at higher 336px resolution for additional 400M samples outperform models with significantly more complexity on popular text-image COCO and Flickr benchmarks. Compared to ALIGN [28] that has approximately twice the number of parameters compared to our DiHT-L/14 model and is trained on 4x bigger data, we improve the performance substantially for all the retrieval benchmarks. Our model also performs better than FILIP [85] which utilizes token-wise similarity to compute the final alignment, thus noticeably increasing the training speed and memory cost. We also outperform Florence [87] on all 4 retrieval benchmarks. Note that Florence [87] utilizes a more recent and powerful Swin-H Vision Transformer architecture [46] with convolutional embeddings [78], and a unified contrastive objective [84]. Our proposed contributions are complementary to FILIP [85] and Florence [87], and we believe additional gains can be achieved when combined. Finally, LiT [89] and BASIC [58] first pre-train model on an large-scale image annotation dataset with cross-entropy before further training with contrastive loss on an image-text dataset. Though this strategy results in state-of-the-art performance on ImageNet1K [65] and image classification benchmarks, it has severe downsides on multi-modal tasks such as cross-modal retrieval. Our ablation in Section 4.2 also confirms this issue. On the other hand, our approach does not suffer from such negative effects.

4.4. Few-shot linear probing

The ideal scenario for leveraging zero-shot recognition models is to warm start the task without training data and then improve the performance (by training a linear probe) via few-shot learning as more and more data is seen. However, in practice, few-shot models perform significantly worse than zero-shot models in the *low-data* regime.

We present an alternate approach to do few-shot classification with prompt-based initialization. The key idea of our approach is to initialize the classifier with the *zero-shot* text prompts for each class, but to also ensure that the final weights do not drift much from the prompt using *projected gradient descent* (PGD) [5]. While few-shot models

Table 6. Comparison with zero-shot state-of-the-art dual-encoder models. px: input image size; #P: model size; #D: training dataset size; #S: total samples processed at training. We evaluate CLIP [62] and OpenCLIP [27] using our codebase, other numbers are copied from respective papers. Grouped models (e.g., ViT-B/32) share same vision and language architecture as our model, following CLIP [62], others have different architectures and we outline the vision one. *FILIP uses token-wise similarity, which is more expensive than global-token similarity and requires adapting the architecture, hence we put it in “Other”.

Method	px	#P	#D	#S	IN	COCO		Flickr	
						T2I	I2T	T2I	I2T
ViT-B/32									
CLIP [62]	224	151M	400M	12.8B	63.4	31.4	49.0	59.5	79.9
OpenCLIP [27]	224	151M	400M	12.8B	62.9	34.8	52.3	61.7	79.2
OpenCLIP [27]	224	151M	2.3B	34B	66.6	39.0	56.7	65.7	81.7
DiHT	224	151M	438M	16B	67.5	40.3	56.3	67.9	83.8
DiHT	224	151M	438M	32B	68.0	40.6	59.3	68.6	84.4
ViT-B/16									
CLIP [62]	224	150M	400M	12.8B	68.4	33.7	51.3	63.3	81.9
OpenCLIP [27]	224	150M	400M	12.8B	67.1	37.8	55.4	65.2	84.1
OpenCLIP [27]	240	150M	400M	12.8B	69.2	40.5	57.8	67.7	85.3
DiHT	224	150M	438M	16B	71.9	43.7	62.0	73.2	89.5
DiHT	224	150M	438M	32B	72.2	43.3	60.3	72.9	89.8
ViT-L/14									
CLIP [62]	224	428M	400M	12.8B	75.6	36.5	54.9	66.1	84.5
CLIP [62]	336	428M	400M	13.2B	76.6	37.7	57.1	68.6	86.6
OpenCLIP [27]	224	428M	400M	12.8B	72.8	42.1	60.1	70.4	86.8
OpenCLIP [27]	224	428M	2.3B	32B	75.2	46.2	64.3	75.4	90.4
DiHT	224	428M	438M	16B	77.0	48.0	65.1	76.7	92.0
DiHT	336	428M	438M	16.4B	77.9	49.3	65.3	78.2	91.1
Other									
ALIGN [28]	289	820M	1.8B	19.7B	76.4	45.6	58.6	75.7	88.6
EfficientNet-L2									
FILIP [85]*	224	428M	340M	10.2B	77.1	45.9	61.3	75.0	89.8
ViT-L/14									
OpenCLIP [27]	224	986M	2.3B	32B	77.9	49.0	67.5	76.8	91.3
ViT-H/14									
Florence [87]	384	893M	900M	31B	83.7	47.2	64.7	76.7	90.9
CoSwin-H									
LiT [89]	288	2.0B	3.6B	18.2B	85.2	41.9	59.3	—	—
ViT-g/14									
BASIC [58]	224	3.1B	6.6B	32.8B	85.7	—	—	—	—
CoAtNet-7									

have been initialized with prompt priors in the past with naive L_2 penalties for weight to prevent catastrophic forgetting [34], these approaches do not improve performance and the model simply ignores the supervision. In contrast, for any target dataset $\mathcal{D}_{\text{target}} = \{(\mathbf{x}_i, y_i)_{i=1}^n\}$, where $\mathbf{x}_i = \phi_{\text{image}}(I_i)$ denotes the image features from the *trained* image tower, we solve the following optimization problem, for some $\delta, \delta_b > 0$:

$$\min_{\|\mathbf{W}\|_2 \leq \delta, \|\mathbf{b}\|_2 \leq \delta_b} \sum_{i=1}^n \mathcal{L}_{\text{CE}}(y_i, \mathbf{x}_i^\top (\mathbf{W} + \mathbf{W}_0) + \mathbf{b}).$$

Here $\mathbf{W}_0 \in \mathbb{R}^{d \times n_c}$ denotes the prompt initialization from the text encoder. To optimize the objective, one can use projected gradient descent [5]. We observe that our approach is able to bridge the gap between zero-shot and 1-shot classification, a common issue in prior linear probe evaluations.

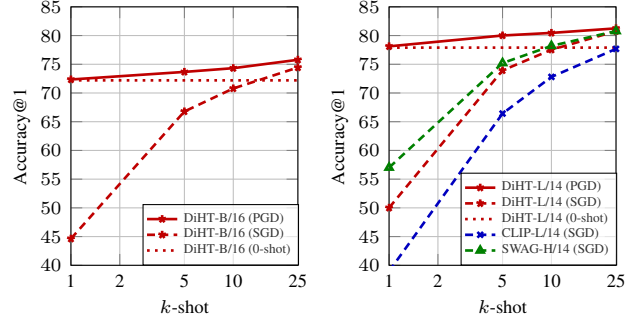


Figure 6. k -shot linear probing performance on ImageNet1K.

Figure 6 presents the full summary of results on the ImageNet1K [65] k -shot classification task. Hyperparameters δ and δ_b for our approach, and weight decay for the baseline approach of training linear probes from scratch are found using grid search. Note that compared to the baseline, our method performs substantially better at very low values of k and maintains the performance continuum from zero-shot to 1-shot, and so on. At large k values, both approaches perform similarly, since there are sufficient data samples to render the zero-shot initialization ineffective. To further showcase the strength of our approach, we also compare our performance with linear probes trained on powerful SWAG [71] models that are especially suited for this task. Note that our approach outperforms the much larger SWAG ViT-H/14 model up to 25-shot classification. We would like to emphasize that this albeit straightforward approach is one of the first to resolve this discontinuity problem between zero-shot and few-shot learning.

5. Conclusion and future work

In this paper, we demonstrate that with careful dataset filtering and simple but effective modeling changes, it is possible to achieve substantial improvements in zero-shot performance on retrieval and classification tasks through large-scale pre-training. Our CAT filtering approach can be applied generically to any large-scale dataset for improved performance with smaller training schedules. Moreover, our concept distillation approach presents a compute and storage efficient way of leveraging very large capacity pre-trained image models for multimodal training. Finally, our simple projected gradient approach covers the crucial performance gap between zero-shot and few-shot learning.

In future, we would like to extend our approach to multi-modal encoder/decoder [1, 10, 41, 83, 86] architectures that although expensive, have better zero-shot performance compared to dual encoders. We also observe that benefits of our hard-negatives loss are less on noisier LAION dataset compared to PMD. It would be interesting to explore how to make it more effective in these very noisy settings. We hope that our improvements and extensive large-scale ablations will further advance the vision-language research.